

Impact of Limited Hall Accommodation on Academic Performance and Wellbeing of NSTU Students (A Machine Learning Approach)

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ABSTRACT

Student accommodation plays a significant role in influencing academic performance and overall wellbeing. Due to the limited capacity of residential halls at Noakhali Science and Technology University (NSTU), many students are required to live in rented accommodation outside the campus. This situation often leads to longer commuting times, increased transportation costs, financial pressure, safety concerns, and reduced study time, which may affect both academic achievement and quality of life.

This study investigates the impact of limited hall accommodation on the academic performance and wellbeing of NSTU students using supervised machine learning techniques. Primary data were collected through a structured Google Forms survey from 116 undergraduate students. The dataset contained 36 features covering demographic information, accommodation conditions, transportation, financial status, academic behavior, and wellbeing indicators.

The collected data were preprocessed through data cleaning, missing value handling, categorical encoding, feature engineering, exploratory data analysis (EDA), and correlation analysis. Six supervised machine learning algorithms-Decision Tree, Random Forest, XGBoost, Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)-were developed and evaluated.

Two prediction tasks were performed. For academic performance prediction using CGPA as the target variable, the Support Vector Machine achieved the highest accuracy of 62.50%**.** For student wellbeing prediction using Living Satisfaction, the highest accuracy for the original five-class problem was 41.67% using both Random Forest and SVM. After regrouping the target into three classes (Low, Medium, and High Satisfaction), K-Nearest Neighbors achieved the best accuracy of 66.67%**.**

The findings indicate that factors such as academic year, attendance, department, internet availability, safety, sleep duration, and accommodation-related conditions contribute to students' academic performance and wellbeing. The study demonstrates that appropriate data preprocessing and model selection can improve predictive performance while providing useful insights for evidence-based decision-making. The results may assist university authorities in improving residential facilities, student support services, and accommodation-related policies.

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1. INTRODUCTION

1.1 Background of the Study

Limited residential accommodation has become a significant challenge for many public universities in Bangladesh. As student enrollment continues to increase, the number of available residential hall seats has not expanded proportionately. Consequently, many students are required to live in rented accommodation outside the campus, often resulting in longer commuting times, higher transportation and living expenses, safety concerns, and reduced access to academic facilities. These challenges may influence students' academic performance, wellbeing, and overall university experience.

Noakhali Science and Technology University (NSTU) faces similar accommodation constraints. Although the university attracts students from across the country, the available hall capacity is insufficient to accommodate all eligible students. As a result, a considerable number of students rely on private housing and commute daily to the campus. Such living arrangements may affect class attendance, study time, sleep patterns, financial stress, and overall living satisfaction.

Understanding how accommodation-related factors influence student outcomes is important for developing effective institutional policies and support services. While previous studies have explored the relationship between accommodation and academic achievement, most have relied on conventional statistical methods or focused on a single outcome. Machine learning offers an alternative approach by identifying complex relationships among multiple variables and enabling predictive analysis of student outcomes.

Machine learning has become an increasingly valuable tool in educational data mining. Algorithms such as Decision Tree, Random Forest, Support Vector Machine (SVM), Logistic Regression, K-Nearest Neighbors (KNN), and XGBoost have demonstrated strong performance in predicting student outcomes, identifying risk factors, and supporting educational decision-making. These algorithms enable researchers to compare multiple predictive models and determine which approach best captures the relationships within a dataset.

Motivated by these opportunities, this study applies supervised machine learning techniques to investigate whether accommodation-related factors influence the academic performance and wellbeing of NSTU students. Primary data were collected through a structured Google Forms questionnaire completed by 116 students from various departments. The collected information includes demographic characteristics, accommodation type, transportation, financial conditions, academic behaviors, and wellbeing indicators. The study then applies a comprehensive machine learning pipeline involving data

preprocessing, exploratory data analysis, feature engineering, model training, evaluation, and feature importance analysis to derive meaningful insights.

Unlike many previous studies that focus solely on academic achievement or student satisfaction, this research simultaneously investigates two important outcomes: academic performance, represented by students' cumulative grade point average (CGPA), and student wellbeing, represented by living satisfaction. This dual-perspective approach provides a more comprehensive understanding of how limited residential facilities may affect students beyond their academic achievements alone.

1.2 Problem Statement

Residential accommodation remains one of the most pressing concerns for students enrolled in public universities throughout Bangladesh. While university admission rates continue to increase annually, the availability of residential halls has not expanded proportionally. Consequently, many students are unable to obtain accommodation within university halls and must arrange private housing outside the campus.

At Noakhali Science and Technology University, this issue has become increasingly significant due to the limited number of available hall seats compared with the total student population. Students residing outside university halls often experience additional financial expenses, longer commuting times, transportation difficulties, safety concerns, and environmental factors that may influence both their academic progress and psychological wellbeing.

Although these challenges are frequently discussed among students, limited empirical evidence exists regarding the actual impact of accommodation conditions on academic performance and wellbeing at NSTU. Furthermore, most previous studies primarily rely on descriptive statistical analyses without leveraging modern machine learning approaches capable of identifying complex relationships among multiple variables.

Without a comprehensive understanding of these relationships, university authorities may find it difficult to prioritize investments in residential facilities, transportation support, or student welfare programs. Therefore, a systematic investigation using machine learning techniques is necessary to determine which accommodation-related factors most strongly influence students' academic performance and wellbeing.

This study seeks to address this research gap by collecting real-world survey data from NSTU students and applying multiple supervised machine learning algorithms to predict academic performance and living satisfaction while identifying the most influential contributing factors.

1.3 Objectives

The primary objective of this study is to investigate the relationship between limited hall accommodation and the academic performance and wellbeing of undergraduate students at Noakhali Science and Technology University (NSTU) using supervised machine learning techniques.

To achieve this objective, the study pursues the following specific objectives:

- To identify the accommodation-related factors associated with students' academic performance and wellbeing.
- To develop and evaluate supervised machine learning models for predicting students' academic performance based on accommodation-related factors.
- To develop and evaluate supervised machine learning models for predicting students' living satisfaction as an indicator of wellbeing.
- To compare the predictive performance of different machine learning algorithms using standard evaluation metrics.
- To identify the most influential features affecting the prediction of academic performance and wellbeing.
- To provide evidence-based recommendations for improving student accommodation policies and support services at NSTU.

1.4 Research Questions

This study attempts to answer the following research questions:

- Does limited hall accommodation significantly influence students' academic performance?
- Which accommodation-related factors have the greatest impact on students' wellbeing?
- Which machine learning algorithm performs best for predicting academic performance?
- Which machine learning algorithm performs best for predicting student wellbeing?
- How do preprocessing techniques such as encoding, feature engineering, and target grouping affect model performance?
- Which variables contribute most significantly to the prediction of academic performance and living satisfaction?

1.5 Scope of the Study

This study investigates the relationship between limited hall accommodation and the academic performance and wellbeing of undergraduate students at Noakhali Science and Technology University

(NSTU). The analysis is based on primary data collected through a structured questionnaire from 116 undergraduate students representing different academic years and departments.

The study focuses on accommodation-related factors, including housing conditions, commuting, transportation, financial status, study environment, internet availability, safety, sleep duration, and other variables that may influence students' academic performance and living satisfaction. Supervised machine learning algorithms are employed to develop predictive models, compare their performance, and identify the most influential factors associated with the target outcomes.

The findings are intended to provide insights that may support university administrators and policymakers in improving accommodation-related services and student support initiatives. However, the study is limited to undergraduate students of NSTU and is based on survey responses collected during the study period. Therefore, the findings may not be directly generalizable to other universities or student populations.

1.6 Significance of the Study

This study contributes to both educational research and practical decision-making in several important ways.

Academically, the research demonstrates the application of machine learning techniques to educational data collected from a Bangladeshi public university. By comparing multiple supervised learning algorithms, the study provides useful insights into their effectiveness in predicting student outcomes.

For university administrators, the findings highlight accommodation-related factors that influence students' academic success and wellbeing. Such information may assist in developing evidence-based policies regarding hall allocation, transportation facilities, financial support, and student welfare initiatives.

For students, the research increases awareness of how living conditions, transportation, study environment, and other accommodation-related factors may affect academic achievement and quality of life.

Finally, this study establishes a foundation for future researchers interested in educational data mining, student analytics, and machine learning applications within higher education institutions in Bangladesh. By expanding the dataset and incorporating additional variables, future studies can further improve predictive accuracy and generate broader institutional insights.

1.7 Brief Methodology

This study adopted a quantitative research approach to investigate the relationship between limited hall accommodation and the academic performance and wellbeing of undergraduate students at Noakhali Science and Technology University (NSTU). Primary data were collected through a structured Google Forms questionnaire completed by 116 undergraduate students from different departments and academic years.

The collected data were preprocessed through data cleaning, missing value treatment, categorical data encoding, and feature engineering to prepare the dataset for machine learning analysis. Exploratory data analysis (EDA) and correlation analysis were also conducted to understand the characteristics of the dataset and the relationships among the variables.

Six supervised machine learning algorithms—Decision Tree, Random Forest, Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and XGBoost—were developed to predict students' academic performance and living satisfaction. Model performance was evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and confusion matrices. The comparative analysis of these models was used to identify the most suitable approach for the prediction tasks.

2. LITERATURE REVIEW

This section reviews previous studies on student accommodation, academic performance, student wellbeing, and the application of machine learning in higher education. It examines the factors associated with students' academic and wellbeing outcomes, summarizes the findings of related research, and identifies gaps that provide the rationale for the present study.

2.1 Student Accommodation

Student accommodation refers to the residential facilities provided or utilized by university students during their academic studies. Accommodation can generally be classified into three categories: university-managed residential halls, privately rented accommodations, and family residences. The quality, accessibility, affordability, and location of accommodation significantly influence students' academic and personal lives.

Residential halls are designed to create an environment that supports learning by providing students with convenient access to classrooms, libraries, laboratories, and extracurricular activities. Living on campus often reduces commuting time, encourages peer interaction, and allows students to participate more actively in university life. Previous studies have suggested that students residing in university accommodation generally experience greater academic engagement and stronger social integration.

However, the demand for residential halls often exceeds the available capacity, particularly in public universities within developing countries. Consequently, many students are compelled to seek alternative housing arrangements outside campus. Such accommodations frequently involve higher living expenses, transportation difficulties, safety concerns, and limited study environments. These factors may negatively influence academic productivity and overall student satisfaction.

In Bangladesh, limited university residential facilities have become a persistent challenge due to increasing student enrollment and insufficient infrastructural expansion. Students living outside campus commonly spend considerable time commuting and incur additional financial burdens. These challenges may reduce study time, increase stress levels, and limit participation in university activities. Therefore, understanding the consequences of accommodation shortages is essential for improving educational policies and student support services.

2.2 Academic Performance and Its Determinants

Academic performance represents the extent to which students achieve educational objectives and is commonly measured through examination scores, Grade Point Average (GPA), or Cumulative Grade

Point Average (CGPA). It serves as one of the primary indicators of student success and institutional effectiveness.

Numerous factors influence academic performance. These factors may be categorized into demographic, socioeconomic, psychological, behavioral, and environmental variables.

Demographic factors include age, gender, academic department, and year of study. Research has shown that students at different academic stages may experience varying academic pressures and responsibilities.

Socioeconomic factors include family income, accommodation costs, transportation expenses, and financial stress. Financial difficulties often force students to balance academic responsibilities with economic challenges, potentially reducing academic performance.

Behavioral factors such as class attendance, study habits, sleep duration, internet accessibility, and extracurricular involvement also play significant roles in determining academic success. High attendance rates and consistent study routines are generally associated with better academic outcomes.

Environmental factors encompass accommodation quality, commuting distance, transportation availability, study environment, and campus accessibility. Students who spend excessive time commuting may have less opportunity for self-study, collaborative learning, and participation in university activities.

Psychological variables including stress, anxiety, academic pressure, and social isolation further influence students' academic achievement. Elevated stress levels may reduce concentration, motivation, and overall learning efficiency.

Because these variables often interact in complex ways, machine learning techniques have become increasingly useful for identifying their combined influence on academic performance.

2.3 Student Wellbeing in University Settings

Student wellbeing has become an important area of research in higher education because of its close relationship with academic success, student engagement, and overall quality of life. Previous studies have reported that students' wellbeing is influenced by a combination of academic, financial, social, and environmental factors. Living conditions, financial stability, social support, and access to university facilities have consistently been identified as important contributors to students' overall wellbeing.

Several studies have found that accommodation-related conditions significantly affect students' living satisfaction and psychological wellbeing. Factors such as housing quality, commuting time, transportation availability, internet access, personal safety, and the availability of suitable study environments have been associated with students' stress levels, life satisfaction, and academic

engagement. Students living in supportive and convenient accommodation generally report higher levels of wellbeing than those facing financial or logistical difficulties in off-campus housing.

Recent studies have also highlighted the growing use of data-driven approaches to identify the factors associated with student wellbeing. Although machine learning techniques have shown promising results in predicting student outcomes, relatively few studies have focused on predicting wellbeing using accommodation-related variables, particularly within the context of Bangladeshi public universities. This gap provides the motivation for the present study.

2.4 Educational Data Mining and Machine Learning

Educational Data Mining (EDM) is an interdisciplinary field that combines education, statistics, data science, and artificial intelligence to analyze educational data and improve learning outcomes. The rapid growth of digital educational systems has generated large volumes of student-related information that can be analyzed using computational techniques.

Machine learning has become one of the most widely adopted approaches within educational data mining because it enables computers to identify patterns, relationships, and predictive models without explicit programming.

Supervised learning algorithms are particularly suitable for predicting student outcomes because they learn from labeled historical data.

Decision Tree algorithms classify data by constructing hierarchical decision rules based on feature values. Their simplicity and interpretability make them popular in educational research.

Random Forest improves prediction performance by combining multiple decision trees through ensemble learning. This approach generally reduces overfitting while increasing predictive stability.

Logistic Regression estimates the probability of class membership using statistical relationships between independent variables and target classes. Although relatively simple, it often performs well on structured educational datasets.

Support Vector Machine (SVM) constructs optimal decision boundaries that maximize class separation. SVM has demonstrated strong performance on classification problems involving high-dimensional datasets.

K-Nearest Neighbors (KNN) classifies observations according to the characteristics of nearby samples within the feature space. Although computationally simple, its performance depends heavily on appropriate scaling and parameter selection.

XGBoost represents an advanced gradient boosting algorithm that has become widely recognized for its predictive performance across numerous machine learning competitions and real-world applications.

Comparing multiple algorithms enables researchers to determine which model best represents the underlying relationships within a particular educational dataset.

2.5 Previous Related Studies

Several international studies have examined the relationship between accommodation conditions and academic outcomes. Research consistently suggests that students residing in supportive residential environments generally demonstrate higher academic engagement, improved attendance, and stronger academic achievement than students experiencing housing instability.

Studies focusing on student wellbeing have similarly found significant associations between accommodation quality, commuting distance, financial stress, and psychological health. Students living closer to campus frequently report lower stress levels, greater participation in university activities, and higher overall satisfaction.

Within the field of educational data mining, numerous researchers have successfully applied machine learning techniques to predict student performance. Decision Trees, Random Forests, Support Vector Machines, Logistic Regression, and XGBoost have all demonstrated promising predictive capabilities depending on dataset characteristics.

In Bangladesh, however, relatively limited research has explored the relationship between university accommodation and student outcomes using machine learning approaches. Existing studies often emphasize descriptive statistics or conventional regression analysis while overlooking the predictive potential of modern artificial intelligence techniques.

Furthermore, few studies simultaneously investigate both academic performance and wellbeing using the same dataset. This limitation highlights the need for more comprehensive investigations integrating educational data mining with social and behavioral variables.

2.6 Research Gap

The reviewed literature demonstrates that student accommodation, academic performance, and wellbeing have been widely investigated in higher education. Previous studies have also shown the potential of machine learning techniques for predicting various student outcomes. However, several important gaps remain.

First, most existing studies have focused on academic performance or wellbeing independently, with relatively few examining both outcomes within a single analytical framework. Second, accommodation-related variables have received comparatively less attention than demographic, behavioral, or academic factors, despite their potential influence on students' educational experiences. Third, most machine learning studies in higher education have been conducted using datasets from institutions outside Bangladesh, limiting the applicability of their findings to the context of Bangladeshi public universities.

Furthermore, comparative evaluations of multiple supervised machine learning algorithms using accommodation-related variables remain limited in the local context. Existing studies also provide limited evidence regarding the relative importance of accommodation-related factors in predicting both academic performance and living satisfaction.

This study addresses these gaps by applying multiple supervised machine learning algorithms to primary survey data collected from undergraduate students at Noakhali Science and Technology University (NSTU). It simultaneously investigates academic performance and living satisfaction, compares the predictive performance of different models, and identifies the accommodation-related factors most strongly associated with these outcomes.

3. RESEARCH METHODOLOGY

This section describes the methodology adopted to investigate the relationship between limited hall accommodation and the academic performance and wellbeing of undergraduate students at Noakhali Science and Technology University (NSTU). It outlines the research design, data collection procedures, dataset preparation, preprocessing techniques, machine learning models, and evaluation methods used in the study.

3.1 Research Design

This study adopted a quantitative research design to investigate the relationship between accommodation-related factors and the academic performance and wellbeing of undergraduate students at Noakhali Science and Technology University (NSTU). A survey-based approach was employed to collect structured data from students, enabling the measurement of demographic, academic, accommodation-related, and wellbeing variables.

Following data collection, the dataset underwent preprocessing to improve its quality and suitability for machine learning analysis. The preprocessing stage included data cleaning, handling missing values, encoding categorical variables, feature engineering, and other necessary transformations.

The processed data were analyzed using multiple supervised machine learning algorithms to develop predictive models for students' academic performance and living satisfaction. The use of several algorithms enabled a comparative evaluation of their predictive capabilities and helped identify the most suitable model for the study. Model performance was assessed using standard classification metrics, including accuracy, precision, recall, F1-score, and confusion matrices, to ensure a comprehensive evaluation of prediction quality.

Overall, the quantitative research design was well suited to the objectives of this study because it supports objective data analysis, facilitates the comparison of multiple predictive models, and enables the identification of accommodation-related factors associated with students' academic performance and wellbeing. The adopted methodology provides a systematic and data-driven framework for understanding the impact of accommodation conditions on student outcomes and offers evidence that may support future research and institutional decision-making.

3.2 Research Methodology Workflow

Figure 1 illustrates the overall workflow adopted in this study. The research began with data collection through a structured questionnaire, followed by data preprocessing, exploratory data analysis, feature engineering, and data balancing. The processed dataset was then used to train and evaluate multiple

supervised machine learning models. Finally, the models were compared based on standard evaluation metrics, and the most influential accommodation-related factors associated with academic performance and living satisfaction were identified.

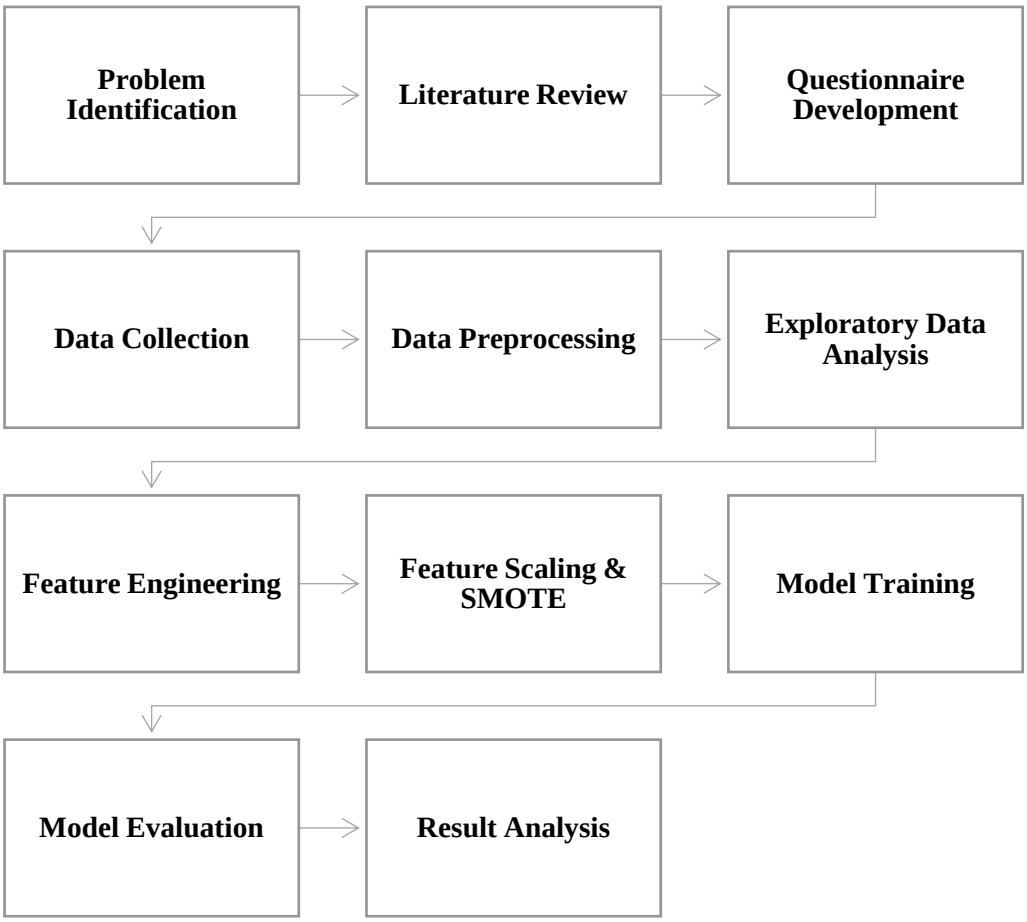


Figure 1: Research Methodology Workflow

3.3 Data Collection

The dataset used in this research was collected through a structured online questionnaire developed using Google Forms.

The questionnaire was designed specifically for this study to gather information regarding students' accommodation experiences and factors potentially influencing their academic performance and wellbeing.

The questionnaire consisted primarily of multiple-choice and Likert-scale questions to ensure consistency and facilitate quantitative analysis.

The survey was distributed among undergraduate students from various departments of Noakhali Science and Technology University through social media platforms and student communication groups.

Participation in the survey was voluntary, and respondents were informed that their identities would remain confidential. No personally identifiable information was collected, ensuring participant privacy and ethical compliance.

3.4 Data Sources

The research utilized primary data, meaning the dataset was collected directly by the researcher rather than obtained from publicly available repositories.

Table 1: Dataset Summary

Attribute	Description
Data Source	Google Forms Survey
Population	NSTU Students
Number of Responses	116
Original Features	36
Data Type	Structured Survey Data
Research Type	Quantitative

3.5 Sampling Technique

The study employed a non-probability convenience sampling technique.

Convenience sampling was selected because participants were recruited based on accessibility and willingness to participate. The questionnaire link was distributed among NSTU students through online communication channels.

Although convenience sampling may introduce sampling bias, it is commonly used in educational research where access to the complete student population is limited.

The sample includes students from multiple departments and academic years, providing reasonable diversity within the collected dataset.

3.6 Description of Dataset

The original dataset contained 116 observations and 36 features representing various aspects of student life. The variables were categorized into several groups.

Demographic Variables

- Gender
- Age Group
- Department
- Academic Year

Accommodation Variables

- Accommodation Type
- Hall Application Status
- Hall Allocation Status
- Room Sharing

Transportation Variables

- Distance from Campus
- Transportation Method
- One-way Commute Time
- Total Daily Commute
- Transportation Cost

Academic Variables

- Attendance
- Missed Classes
- Study Environment
- Study Difficulty
- Academic Pressure
- CGPA

Wellbeing Variables

- Stress
- Anxiety
- Sleep Hours

- Living Satisfaction
- Isolation
- Safety
- Safety Impact

Financial Variables

- Living Cost
- Financial Stress
- Cost Affect

3.7 Data Preprocessing

Raw survey data cannot be directly used for machine learning because it often contains inconsistencies, missing values, and categorical variables.

Therefore, several preprocessing steps were performed before model development. The preprocessing pipeline consisted of:

- Column renaming
- Data cleaning
- Handling missing values
- Correcting inconsistent categories
- Encoding categorical variables
- Feature engineering

3.7.1 Renaming Columns

The original Google Forms column names contained spaces and special characters that were unsuitable for programming. Therefore, all variables were renamed using lowercase letters and underscores.

For example, "Living Satisfaction" became "living_sat", "Academic Pressure" became "acad_pressure", and "Transportation Cost" became "transport_cost". This improved code readability and compatibility with Python libraries.

3.7.2 Handling Missing Values

Missing values were identified using the isnull() function in Pandas. Categorical missing values were replaced using the mode (most frequent value). Numerical missing values were imputed using appropriate statistical measures.Only one missing value was found in the Distance variable after

preprocessing, which was replaced using the column mean. This ensured that no missing values remained before model training.

3.7.3 Handling Inconsistent Categories

Several categorical variables contained inconsistent spellings and naming conventions. For example, "Software Engineering" and "SWES" represented the same department.

Similarly, variations in department names and categorical responses were standardized to maintain consistency. This prevented duplicate categories during encoding and improved model performance.

3.7.4 Encoding Categorical Variables

Machine learning algorithms cannot process textual data directly. Therefore, categorical variables were converted into numerical representations using Label Encoding.

Binary variables such as Yes/No were encoded as Yes = 1 and No = 0. Examples include Hall Apply, Hall Status, and Missed Class.

Multi-category variables such as Department, Gender, Accommodation Type, and Transportation Method were encoded into numerical labels for efficient model training.

3.7.5 Feature Engineering

Several variables were transformed to improve machine learning performance. The Age Group variable was converted from categorical ranges into approximate numerical values (e.g., 17–19 → 18, 20–22 → 21, 23–25 → 24, Above 25 → 27).

Additionally, the Living Satisfaction variable was grouped from five classes into three broader categories for the second prediction task to reduce class imbalance and improve predictive accuracy.

3.7.6 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the structure and distribution of the dataset before model training. The following visualizations were created: CGPA Distribution, Living Satisfaction Distribution, Accommodation Type Distribution, Department Distribution, and Gender Distribution.

Descriptive statistics were also computed to summarize the dataset. EDA helped identify class imbalance and guided later preprocessing decisions such as regrouping Living Satisfaction categories.

3.8 Correlation Analysis

Correlation analysis was performed to examine the relationships among numerical variables in the dataset before developing predictive models. Understanding these relationships helps identify features that may have stronger influence on the target variables and assists in feature selection.

A correlation matrix was generated using the Pearson correlation coefficient, followed by a heatmap visualization created with the Matplotlib library. The heatmap provided a graphical representation of positive and negative relationships among variables, making it easier to identify highly correlated features.

The analysis revealed that variables related to attendance, academic year, internet availability, safety, study environment, and financial stress exhibited meaningful relationships with the target variables. However, no pair of independent variables showed extremely high correlation, indicating that severe multicollinearity was not present in the dataset. Therefore, all relevant variables were retained for model training.

Correlation analysis also supported the interpretation of machine learning results by providing an initial understanding of how accommodation-related factors interacted with academic performance and student wellbeing.

3.9 Train-Test Split

To evaluate the predictive performance of the machine learning models objectively, the dataset was divided into separate training and testing subsets using the Train-Test Split technique provided by the Scikit-learn library.

An 80:20 split was adopted, where 80% of the observations were used to train the models and the remaining 20% were reserved for testing. This approach allows the models to learn patterns from the training data while evaluating their performance on previously unseen data.

A fixed random state of 42 was used to ensure reproducibility of the experiments. Using the same random seed guarantees that the dataset is divided consistently each time the program is executed.

The train-test split prevents data leakage and provides a realistic estimate of how well the trained models generalize to new observations.

3.10 Feature Scaling

Some machine learning algorithms are sensitive to differences in feature magnitude. Variables measured on larger numerical scales may dominate those measured on smaller scales, leading to biased model performance.

To address this issue, StandardScaler was applied to standardize the numerical features. Standardization transforms each feature by subtracting its mean and dividing by its standard deviation, resulting in a distribution with a mean of zero and a standard deviation of one.

Feature scaling was applied only to algorithms that require normalized input data, including Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN).

Tree-based algorithms such as Decision Tree, Random Forest, and XGBoost do not require feature scaling because they split data based on feature thresholds rather than distance calculations.

3.11 Handling Class Imbalance Using SMOTE

Class imbalance is a common issue in classification problems where one or more target classes contain significantly fewer observations than others. Such imbalance may cause machine learning models to become biased toward the majority class, reducing predictive performance for minority classes.

During the CGPA prediction task, the target variable exhibited an imbalanced distribution, with one class containing considerably fewer samples than the others.

To reduce this imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was applied to the training dataset. SMOTE generates synthetic samples for minority classes by interpolating between existing observations, thereby balancing the class distribution without simply duplicating data.

Importantly, SMOTE was applied only to the training data after the train-test split. The testing dataset remained unchanged to ensure an unbiased evaluation of model performance.

Applying SMOTE improved class balance and allowed the models to learn more representative decision boundaries for minority classes.

3.12 Machine Learning Models

Six supervised machine learning algorithms were implemented and compared in this research. Each model was selected based on its popularity, effectiveness in classification tasks, and suitability for structured tabular datasets.

3.12.1 Decision Tree

Decision Tree is a supervised learning algorithm that recursively partitions the dataset into smaller subsets based on feature values. It is simple to interpret and provides clear decision rules. In this study, the Decision Tree classifier achieved the second-highest prediction accuracy for CGPA classification, behind Support Vector Machine.

3.12.2 Random Forest

Random Forest is an ensemble learning technique that combines multiple decision trees to improve predictive accuracy and reduce overfitting. Each tree is trained on a random subset of the data, and the final prediction is determined by majority voting.

3.12.3 XGBoost

Extreme Gradient Boosting (XGBoost) is a powerful boosting algorithm that sequentially builds decision trees, with each tree correcting the errors made by previous ones. XGBoost is known for its high predictive performance and computational efficiency.

3.12.4 Logistic Regression

Logistic Regression is a statistical classification algorithm that estimates the probability of an observation belonging to a particular class. Despite its simplicity, it often performs exceptionally well on structured datasets, though in this study it did not outperform the top-ranked models on either prediction task.

3.12.5 Support Vector Machine (SVM)

Support Vector Machine constructs an optimal hyperplane that maximizes the separation between classes. It is particularly effective for high-dimensional datasets and, in this study, tied with Random Forest for the best performance on the original five-class living satisfaction target, while also achieving the highest accuracy for CGPA prediction.

3.12.6 K-Nearest Neighbors (KNN)

K-Nearest Neighbors classifies observations based on the majority class of their nearest neighboring data points. The algorithm relies on distance calculations and therefore requires standardized features before training. In this study, KNN achieved the highest accuracy for the three-class Living Satisfaction prediction task.

Table 2: Machine Learning Models Used

Model	Purpose	Scaling Required
Decision Tree	Classification	No
Random Forest	Classification	No
XGBoost	Classification	No
Logistic Regression	Classification	Yes
SVM	Classification	Yes
KNN	Classification	Yes

3.13 Model Evaluation Metrics

To compare the performance of different machine learning algorithms, several standard classification metrics were used.

Accuracy

Accuracy measures the proportion of correctly classified observations among all observations. It provides an overall indication of model performance: Accuracy = (TP + TN) / (TP + TN + FP + FN).

Precision

Precision measures the proportion of correctly predicted positive observations among all observations predicted as positive. High precision indicates that the model makes relatively few false positive predictions.

Recall

Recall measures the proportion of actual positive observations that are correctly identified by the model. High recall indicates that the model successfully identifies most positive instances.

F1-Score

The F1-score represents the harmonic mean of precision and recall. It is particularly useful when dealing with imbalanced datasets because it balances both false positives and false negatives.

Confusion Matrix

A confusion matrix provides a detailed summary of classification results by displaying the number of correct and incorrect predictions for each class. The confusion matrix helps identify which classes are frequently misclassified and supports detailed performance analysis.

3.14 Software and Tools

The implementation of this research was carried out using open-source software and libraries widely adopted within the machine learning community.

Table 3: Software and Tools

Software / Tool	Purpose
Python	Programming Language
Jupyter Notebook	Development Environment
Pandas	Data Manipulation

Software / Tool	Purpose
NumPy	Numerical Computation
Matplotlib	Data Visualization
Scikit-learn	Machine Learning
XGBoost	Gradient Boosting Algorithm
Imbalanced-learn	SMOTE Implementation
Google Forms	Data Collection
Microsoft Excel	Initial Data Review
Microsoft Word	Report Preparation

3.15 Ethical Considerations

Ethical considerations were maintained throughout the research process. Participation in the survey was entirely voluntary, and respondents were informed about the purpose of the study before submitting their responses. No personally identifiable information, such as names, student identification numbers, or contact details, was collected.

The collected data were used exclusively for academic research purposes. Responses were analyzed in aggregate form, ensuring participant confidentiality and privacy. The dataset was stored securely, and no information was shared with unauthorized individuals.

4. FINDINGS AND RESULTS

This section presents the findings obtained from the collected dataset after preprocessing, exploratory data analysis, feature engineering, and machine learning model development. The objective of this chapter is to analyze the characteristics of the dataset, evaluate the performance of different classification algorithms, identify the most influential predictors, and interpret the results in relation to the research objectives.

Two prediction tasks were performed in this study. The first focused on predicting students' academic performance (CGPA), while the second investigated student wellbeing (Living Satisfaction). Multiple supervised machine learning algorithms were trained and compared to determine the most suitable model for each prediction task.

4.1 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the dataset before applying machine learning algorithms. Various statistical summaries and graphical visualizations were generated to examine the distributions of the variables.

The analysis revealed that students in the dataset represented multiple academic departments and years of study, with varying accommodation types including university halls, rented messes or flats, living with family, and other arrangements. Accommodation-related factors such as commuting distance, transportation method, transportation cost, room-sharing arrangements, and living expenses showed noticeable variation among respondents.

The CGPA target variable exhibited an imbalanced distribution, with a relatively small number of students belonging to the lowest CGPA category. This imbalance motivated the application of SMOTE during model development.

Similarly, the original five-level Living Satisfaction variable showed unequal class frequencies. To address this issue, an additional experiment grouped the satisfaction levels into three broader categories: Low, Medium, and High Satisfaction. This grouping improved class balance and enhanced predictive performance.

Figures 2 through 4 illustrate the accommodation type, living satisfaction, and CGPA distributions of the sampled students.

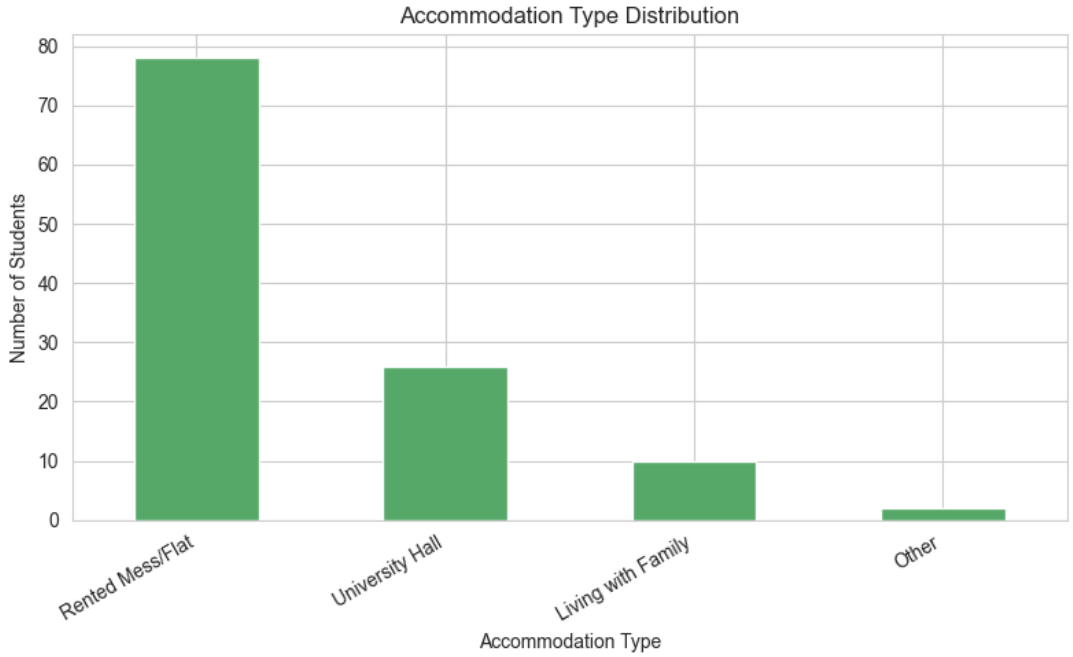


Figure 2: Accommodation Type Distribution

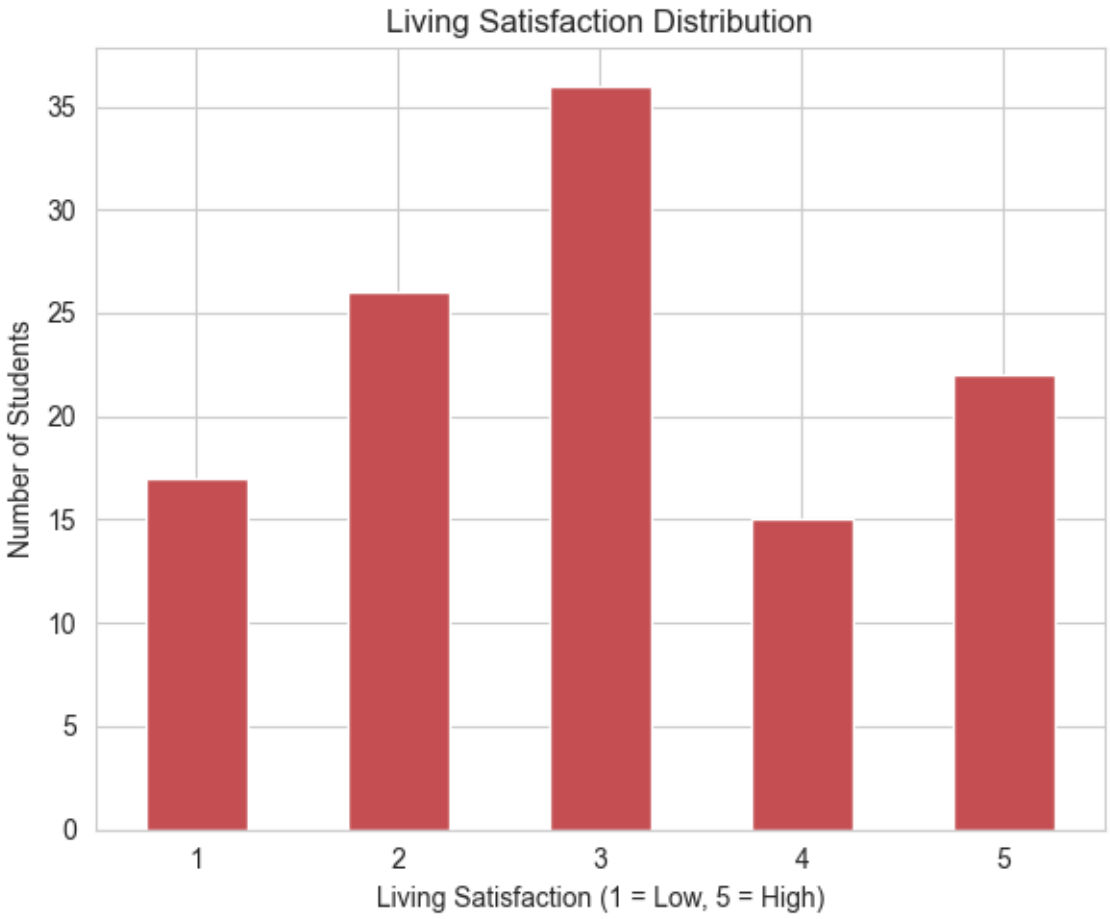


Figure 3: Living Satisfaction Distribution

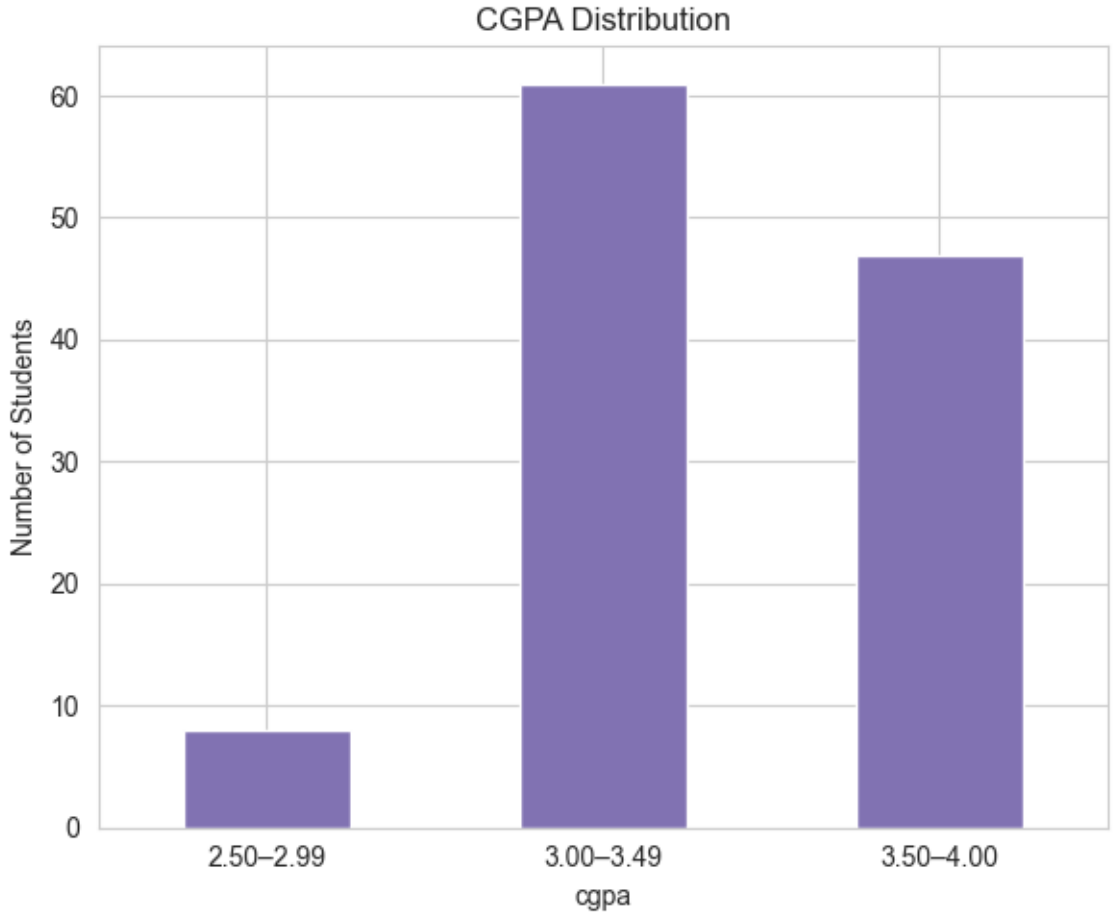


Figure 4: CGPA Distribution

4.2 Correlation Analysis

A Pearson correlation matrix was generated to investigate the relationships among numerical variables in the dataset. The heatmap provided an overview of positive and negative correlations between different features.

The analysis suggested that variables related to academic year, attendance, internet availability, study environment, safety, financial stress, and living satisfaction were associated with student outcomes. No severe multicollinearity was observed among the independent variables, indicating that the selected features could be retained for model development without significantly affecting model stability.

The correlation analysis also supported later feature importance results, as several highly influential variables identified by the machine learning models corresponded with variables showing stronger associations in the correlation matrix.

Figure 5 presents the full correlation heatmap of the numerical and encoded categorical variables.

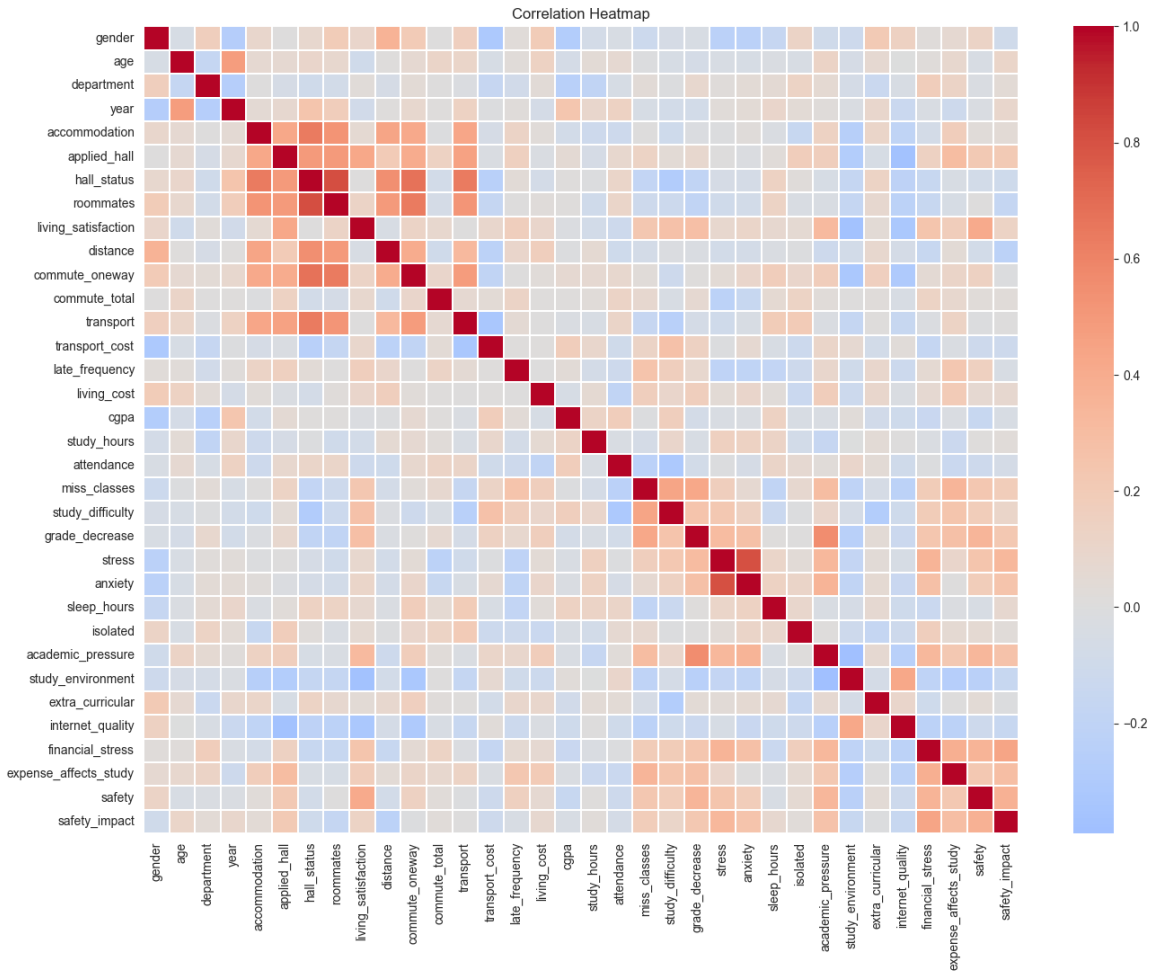


Figure 5: Correlation Heatmap

4.3 Feature Importance Analysis

To identify the variables contributing most significantly to prediction performance, feature importance analysis was conducted using the Random Forest classifier.

The analysis indicated that academic year was the most influential predictor, followed by department, attendance, age group, internet availability, sleep hours, safety, and living satisfaction.

Table 4: Top Features Influencing Prediction

Rank	Feature	Importance
1	Academic Year	0.0916
2	Department	0.0707
3	Gender	0.0642

Rank	Feature	Importance
4	Age Group	0.0629
5	Safety	0.0486
6	Internet Availability	0.0466
7	Transportation Method	0.0465
8	Grade Decrease	0.0461
9	Living Satisfaction	0.0298
10	Safety Impact	0.0285

The importance scores indicate that accommodation-related factors alone were not the only determinants of student outcomes. Instead, academic behavior, access to resources, and personal wellbeing collectively influenced predictive performance.

Figure 6 visualizes the top ten features ranked by Random Forest importance for the CGPA prediction task.

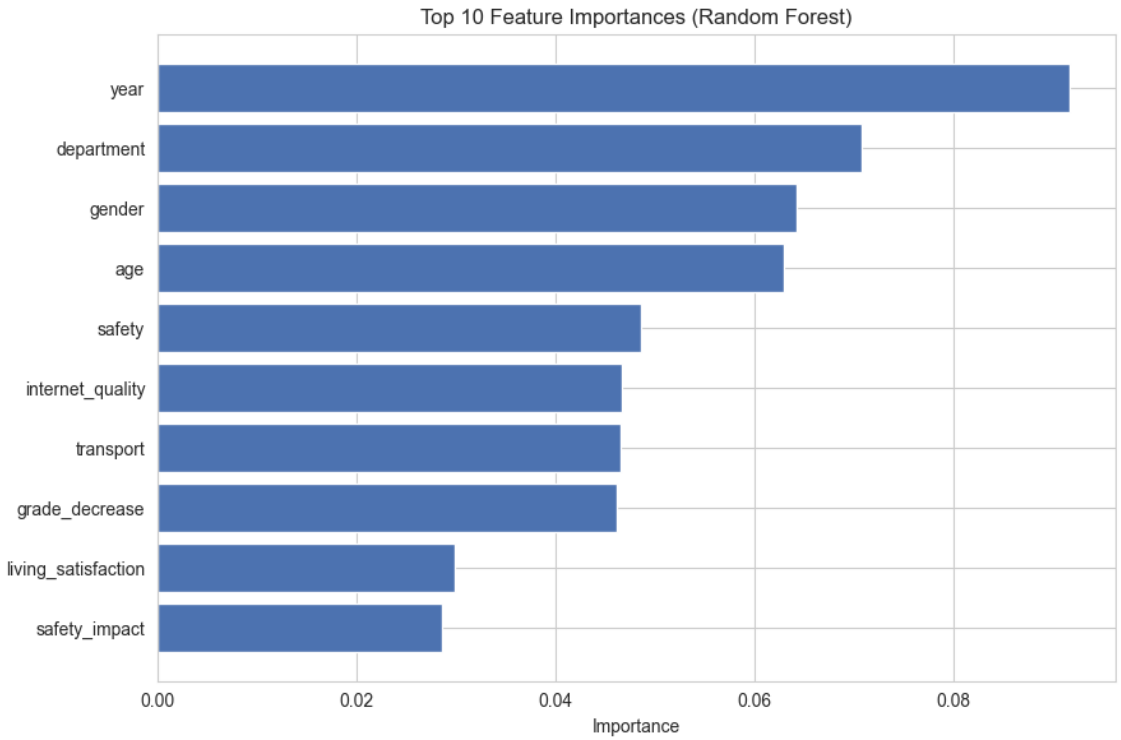


Figure 6: Top 10 Feature Importances (Random Forest)

4.4 Academic Performance Prediction (CGPA)

The first machine learning experiment aimed to predict students' academic performance using CGPA as the target variable. Six supervised learning algorithms were trained and evaluated.

Table 5: Model Performance for CGPA Prediction

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
Support Vector Machine	62.50%	63.62%	62.50%	62.10%
Decision Tree	58.33%	59.31%	58.33%	58.33%
Random Forest	41.67%	41.03%	41.67%	40.84%
XGBoost	41.67%	41.67%	41.67%	40.00%
Logistic Regression	33.33%	36.11%	33.33%	34.20%
K-Nearest Neighbors	33.33%	40.95%	33.33%	33.27%

The results indicate that the Support Vector Machine achieved the highest classification accuracy of 62.50%, outperforming all other evaluated models, with the Decision Tree classifier close behind at 58.33%.

The strong performance of the Support Vector Machine suggests that the relationships between accommodation-related factors and academic performance were reasonably well captured by an optimal separating boundary between CGPA classes. The Decision Tree's close second-place accuracy, combined with its interpretability, still makes it a valuable model for explaining the influence of different variables on students' academic outcomes.

Figure 7 presents the confusion matrix for the Support Vector Machine on the CGPA prediction task.

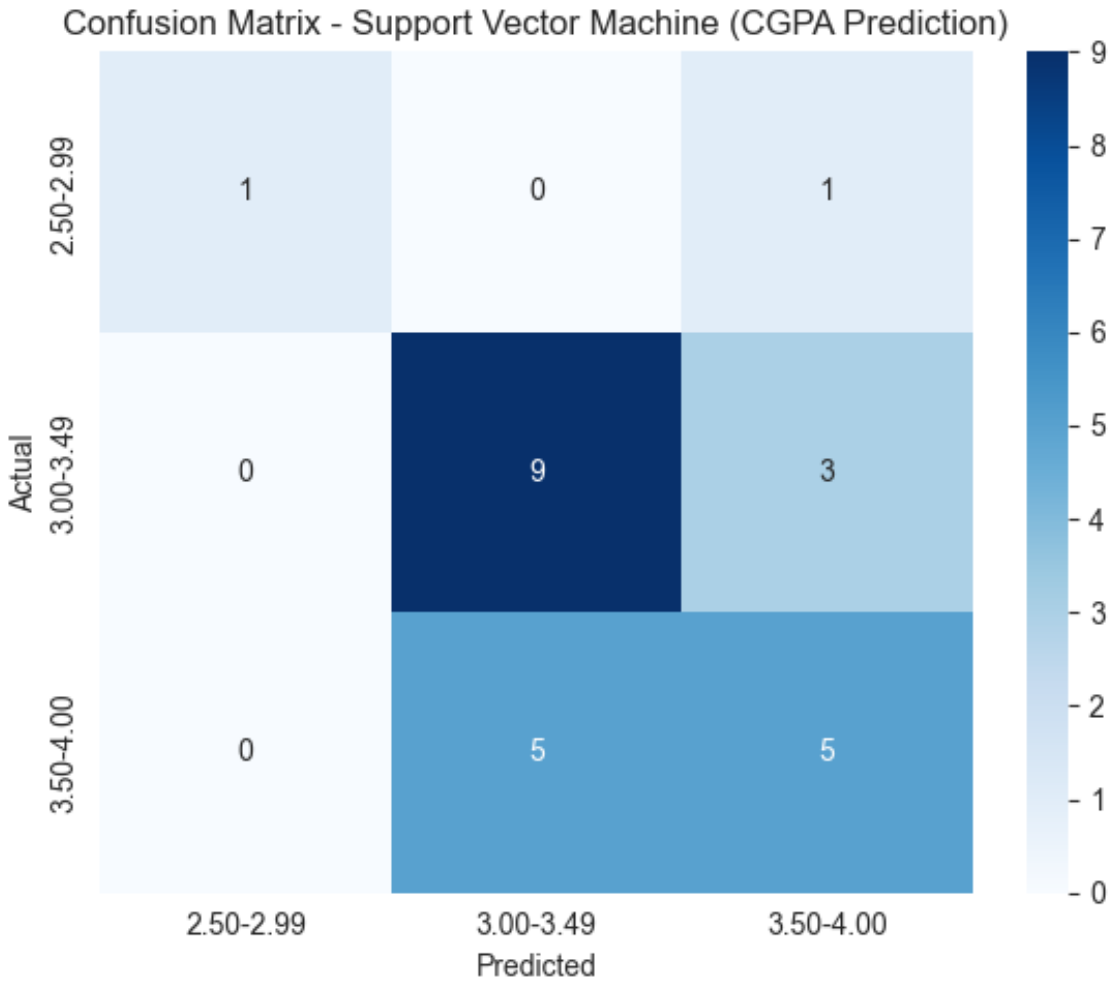


Figure 7: Confusion Matrix - Support Vector Machine (CGPA Prediction)

4.5 Living Satisfaction Prediction (Five-Class Classification)

The second prediction task focused on student wellbeing using the original five-level Living Satisfaction variable.

Table 6: Model Performance for Five-Class Living Satisfaction Prediction

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
Random Forest	41.67%	41.32%	41.67%	40.57%
Support Vector Machine	41.67%	49.17%	41.67%	38.92%
XGBoost	29.17%	24.90%	29.17%	25.47%
Logistic Regression	25.00%	31.71%	25.00%	25.97%

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
Decision Tree	20.83%	16.89%	20.83%	18.33%
K-Nearest Neighbors	20.83%	15.21%	20.83%	16.94%

For the original five-class Living Satisfaction problem, Random Forest and Support Vector Machine tied for the best accuracy at 41.67%, while the remaining models ranged from 20.83% to 29.17%.

The comparatively lower performance is expected because predicting five distinct satisfaction levels is a more challenging multi-class classification problem, especially with a relatively small dataset.

Figure 8 presents the confusion matrix for Random Forest on the five-class Living Satisfaction prediction task.

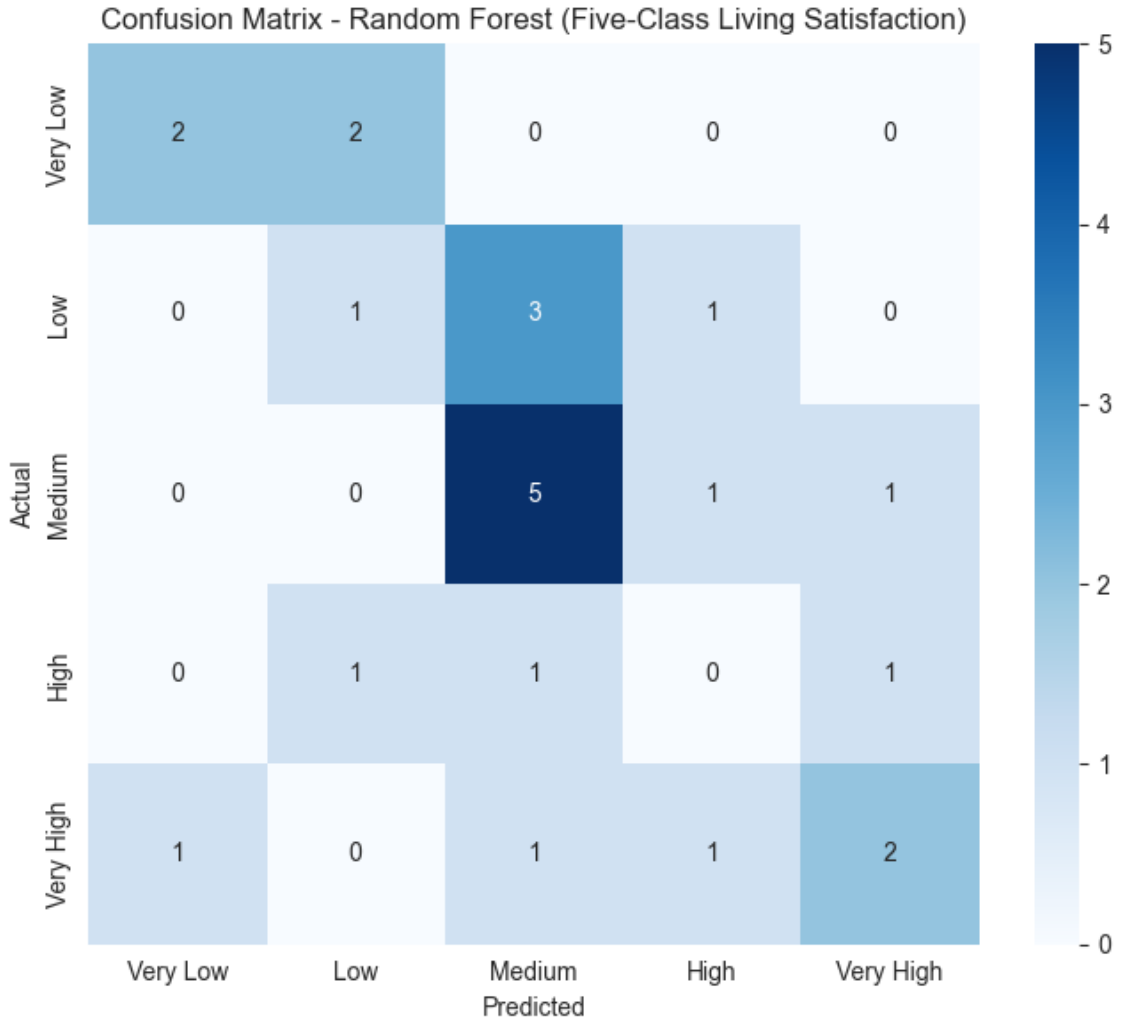


Figure 8: Confusion Matrix - Random Forest (Five-Class Living Satisfaction)

4.6 Living Satisfaction Prediction (Three-Class Classification)

To reduce class imbalance and simplify the prediction task, the original five satisfaction levels were grouped into three categories: Low Satisfaction, Medium Satisfaction, and High Satisfaction. This preprocessing step produced a more balanced target distribution and improved model performance.

Table 7: Model Performance for Three-Class Living Satisfaction Prediction

Model	Accuracy	Precision	Recall	F1 Score
K-Nearest Neighbors	66.67%	65.67%	66.67%	65.68%
Support Vector Machine	58.33%	56.04%	58.33%	56.75%
Decision Tree	50.00%	52.78%	50.00%	50.27%
Random Forest	45.83%	45.37%	45.83%	45.49%
Logistic Regression	41.67%	42.41%	41.67%	41.94%
XGBoost	41.67%	44.84%	41.67%	42.71%

The results demonstrate that K-Nearest Neighbors achieved the highest accuracy of 66.67%, with Support Vector Machine close behind at 58.33%. This improvement confirms that reducing the number of target classes helped the models learn more meaningful decision boundaries.

Figure 9 presents the confusion matrix for K-Nearest Neighbors on the three-class Living Satisfaction prediction task.

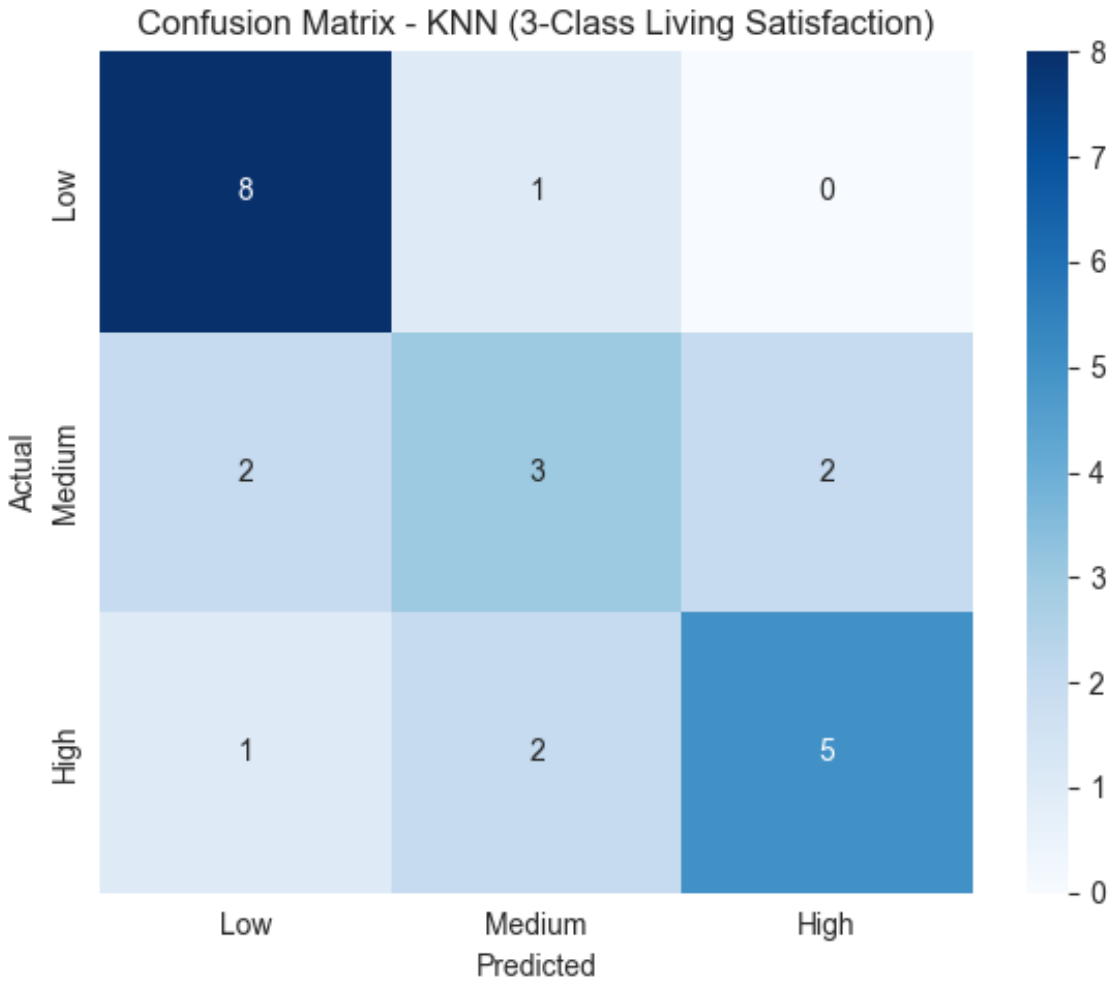


Figure 9: Confusion Matrix - K-Nearest Neighbors (Three-Class Living Satisfaction)

4.7 Comparative Analysis of Models

The experiments demonstrate that different machine learning algorithms perform differently depending on the prediction task.

For predicting academic performance, the Support Vector Machine achieved the highest accuracy, suggesting that an optimal separating boundary captured the relationships among accommodation-related variables reasonably well.

For the original five-class wellbeing prediction, Support Vector Machine performed best, indicating its effectiveness in handling multi-class classification problems with complex decision boundaries.

After regrouping Living Satisfaction into three categories, K-Nearest Neighbors became the best-performing model. This result illustrates the importance of target engineering and class balancing in improving predictive performance.

Overall, the findings suggest that no single machine learning algorithm is universally optimal. Model performance depends on the characteristics of the target variable, feature distribution, and class balance.

Figure 10 summarizes and compares the accuracy of all six models across the three prediction tasks.

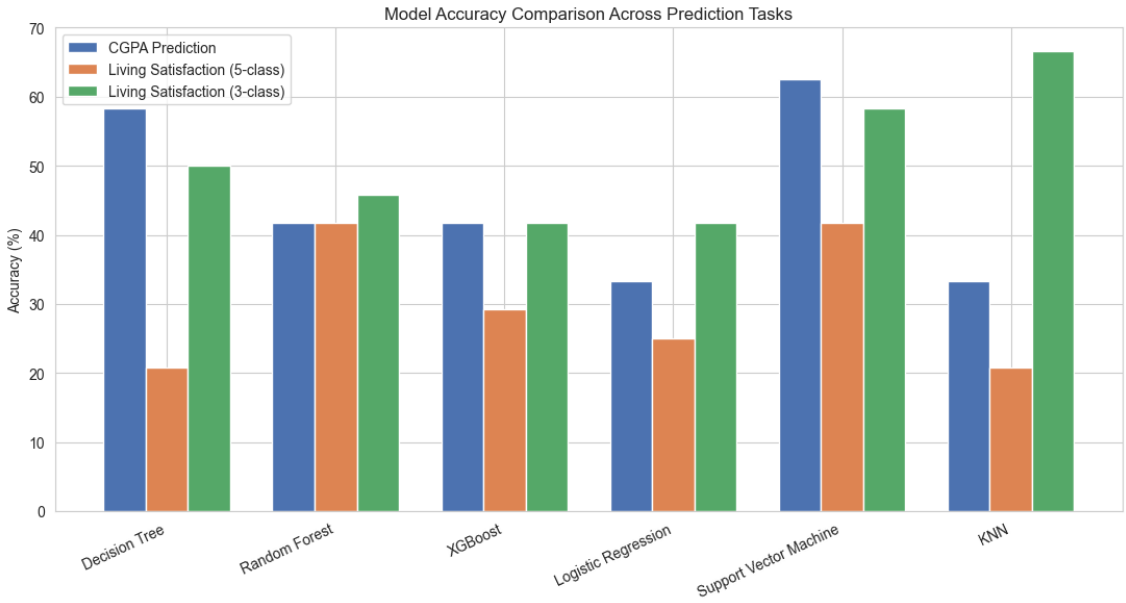


Figure 10: Model Accuracy Comparison Across Prediction Tasks

5. DISCUSSION

The primary objective of this study was to investigate whether limited hall accommodation influences the academic performance and wellbeing of undergraduate students at Noakhali Science and Technology University (NSTU). To achieve this objective, multiple supervised machine learning models were developed and evaluated using primary survey data collected from 116 students.

This section discusses the experimental findings presented in Chapter 4, interprets the model performance, compares the results with previous studies, identifies the most influential predictors, and highlights the implications of the findings for students and university administrators.

5.1 Discussion of Dataset Characteristics

The dataset represented students from different academic departments, years of study, and accommodation types. Respondents included students residing in university halls, rented messes or flats, family residences, and other living arrangements. This diversity allowed the models to capture a wide range of accommodation experiences and their possible effects on student outcomes.

However, the dataset also exhibited class imbalance, particularly for the CGPA target variable. The lowest CGPA category contained significantly fewer observations than the higher categories. Such imbalance can negatively affect model learning by causing algorithms to favor majority classes.

Similarly, the original five-level Living Satisfaction target displayed unequal class frequencies. This imbalance motivated the additional experiment in which the satisfaction levels were grouped into three broader categories.

Although the dataset consisted of only 116 responses, it provided sufficient information to demonstrate an end-to-end machine learning workflow and explore meaningful relationships between accommodation-related variables and student outcomes.

5.2 Discussion of Data Preprocessing

Data preprocessing played a critical role in improving the quality of the dataset before model training. Several preprocessing operations were performed, including:

- Renaming columns
- Correcting inconsistent categories
- Handling missing values
- Label encoding categorical variables
- Feature engineering
- Standardization

- SMOTE for class balancing

These preprocessing techniques ensured that the machine learning algorithms received clean and consistent numerical input.

An important preprocessing decision involved transforming the original Age Group categories into numerical representations, allowing algorithms to recognize age-related ordering.

Similarly, grouping the original five-level Living Satisfaction target into three categories reduced class imbalance and improved prediction performance.

The improvement observed after regrouping demonstrates that appropriate feature engineering and target transformation can substantially influence machine learning accuracy.

5.3 Discussion of Feature Importance

Feature importance analysis revealed that several variables contributed more significantly than others to predicting student outcomes. The most influential features included:

- Academic Year
- Department
- Attendance
- Age Group
- Internet Availability
- Sleep Hours
- Safety
- Living Satisfaction
- Extracurricular Activities

Among these variables, Academic Year achieved the highest importance score, suggesting that students at different stages of their academic journey experience varying academic pressures and accommodation-related challenges.

Attendance emerged as another highly influential factor. Students who attend classes regularly generally demonstrate stronger academic performance because they receive continuous instruction and participate more actively in classroom activities.

Internet availability also ranked among the most important predictors. Access to reliable internet has become essential for university students, supporting online learning, assignment completion, research activities, and communication with instructors.

Safety and sleep duration similarly contributed to prediction performance. Students living in secure environments with healthier sleeping patterns are more likely to maintain better concentration, lower stress levels, and improved academic outcomes.

These findings suggest that academic success depends not only on accommodation type but also on a combination of educational, behavioral, environmental, and psychological factors.

5.4 Discussion of Academic Performance Prediction

The first prediction task focused on academic performance using CGPA as the target variable. Among the six evaluated algorithms, the Support Vector Machine achieved the highest accuracy of 62.50%, with the Decision Tree classifier close behind at 58.33%. Several factors may explain this result.

Decision Trees effectively capture nonlinear relationships between variables without requiring complex mathematical assumptions. They also handle mixed numerical and categorical data efficiently, making them particularly suitable for structured survey datasets.

Unlike linear models, Decision Trees can naturally identify interaction effects among accommodation conditions, attendance, transportation, and demographic characteristics.

The comparatively lower performance of Random Forest and XGBoost may be attributed to the relatively small dataset. Ensemble algorithms generally benefit from larger datasets containing hundreds or thousands of observations. With only 116 samples, the additional complexity of ensemble methods did not translate into improved predictive performance.

These findings indicate that simpler models may outperform more sophisticated algorithms when datasets are relatively small.

5.5 Discussion of Wellbeing Prediction

The second prediction task investigated student wellbeing using Living Satisfaction as the target variable. Initially, the original five-class target was used. Among all evaluated algorithms, Support Vector Machine achieved the highest accuracy of 41.67%.

Predicting five satisfaction levels proved considerably more difficult than predicting CGPA because the target contained greater class overlap and higher subjectivity. Living satisfaction depends on numerous personal perceptions that may not be fully captured by the available survey variables.

To improve predictive performance, the five original satisfaction categories were grouped into three broader classes representing Low, Medium, and High Satisfaction. This transformation reduced class imbalance and simplified the classification problem.

Following regrouping, K-Nearest Neighbors achieved the highest accuracy of 66.67%. This improvement demonstrates that thoughtful target engineering can significantly enhance predictive performance while maintaining meaningful interpretation.

5.6 Comparison with Previous Studies

The findings of this research are generally consistent with previous studies examining student accommodation and academic performance.

Earlier educational studies have reported that accommodation quality, commuting distance, transportation, financial stress, and study environment influence students' academic achievement and psychological wellbeing.

Similarly, educational data mining research has demonstrated the effectiveness of Decision Trees, Random Forests, Logistic Regression, Support Vector Machines, and XGBoost for predicting educational outcomes.

However, unlike many previous studies relying primarily on descriptive statistics or traditional regression analysis, the present research compared six different machine learning algorithms using real-world survey data collected directly from NSTU students.

Furthermore, this study simultaneously investigated academic performance and student wellbeing within the same dataset, providing a broader understanding of accommodation-related challenges.

5.7 Practical Implications

The findings of this research have practical implications for university administrators and policymakers.

The results suggest that improving residential facilities alone may not fully address students' academic challenges. Other factors, including internet accessibility, study environment, transportation, safety, and attendance, also contribute significantly to student success.

University authorities may therefore consider:

- Expanding residential hall capacity.
- Improving campus transportation facilities.
- Enhancing internet accessibility in residential areas.
- Providing safer accommodation environments.
- Offering academic and psychological support programs for students living outside campus.

These interventions may contribute to improved academic performance and overall student wellbeing.

5.8 Limitations of the Study

Although this research produced meaningful findings, several limitations should be acknowledged.

First, the dataset consisted of 116 student responses, which limits the generalizability of the findings. Larger datasets typically enable machine learning models to learn more representative patterns and improve predictive accuracy.

Second, the study relied on self-reported survey responses. Participants' perceptions may introduce response bias or inaccuracies.

Third, the dataset represents only students from Noakhali Science and Technology University. Consequently, the findings may not directly apply to other universities with different residential facilities, academic systems, or student populations.

Finally, the study considered a limited number of variables. Additional factors such as family income, parental education, mental health assessments, previous academic records, and extracurricular involvement could further improve predictive performance.

6. CONCLUSION

This research investigated the impact of limited hall accommodation on the academic performance and wellbeing of undergraduate students at Noakhali Science and Technology University (NSTU) using supervised machine learning techniques. The study combined primary data collection, comprehensive data preprocessing, exploratory data analysis, and predictive modeling to identify the accommodation-related factors that influence students' educational outcomes.

A total of 116 student responses were collected through a structured Google Forms questionnaire. The collected data represented students from different departments, academic years, accommodation types, and demographic backgrounds. Following preprocessing, six supervised machine learning algorithms were trained and evaluated to predict both academic performance (CGPA) and student wellbeing (Living Satisfaction).

The findings provide valuable insights into the role of accommodation conditions in shaping students' academic and personal experiences.

6.1 Summary of Findings

The research successfully developed an end-to-end machine learning pipeline beginning with survey design and ending with predictive model evaluation. The collected dataset underwent several preprocessing steps, including cleaning, handling missing values, correcting inconsistent categories, encoding categorical variables, feature engineering, feature scaling, and class balancing using SMOTE.

Exploratory Data Analysis (EDA) and correlation analysis provided a comprehensive understanding of the dataset before model development. Feature importance analysis identified the variables that contributed most significantly to prediction performance.

Among the six evaluated machine learning algorithms, Support Vector Machine achieved the highest classification accuracy for predicting academic performance (CGPA), with an accuracy of 62.50%. This indicates that Support Vector Machine effectively captured the relationships between accommodation-related variables and students' academic outcomes within the collected dataset.

For predicting the original five-class Living Satisfaction target, Support Vector Machine (SVM) achieved the highest accuracy of 41.67%. Although the performance was moderate, it reflected the complexity of predicting subjective wellbeing using a relatively small dataset.

To improve prediction performance, the original five satisfaction levels were grouped into three broader categories representing Low, Medium, and High Satisfaction. After regrouping, K-Nearest

Neighbors produced the highest classification accuracy of 66.67%, demonstrating that target engineering and improved class balance significantly enhanced model performance.

Feature importance analysis revealed that academic year, department, attendance, age group, internet availability, sleep hours, safety, and living satisfaction were among the most influential predictors. These findings suggest that academic success and wellbeing are influenced by multiple interconnected factors rather than accommodation status alone.

Overall, the results demonstrate that machine learning techniques can effectively analyze educational survey data and identify meaningful patterns that support evidence-based decision-making.

6.2 Achievement of Research Objectives

The study successfully achieved all of its proposed objectives.

The first objective, which involved collecting primary data related to accommodation conditions and student experiences, was accomplished through a structured Google Forms survey involving 116 undergraduate students from NSTU.

The second objective of preparing a machine-learning-ready dataset was achieved through systematic preprocessing, including data cleaning, missing value treatment, categorical encoding, and feature engineering.

The third objective of exploring the dataset through statistical summaries and visualizations was fulfilled using Exploratory Data Analysis and correlation analysis.

The fourth objective of developing predictive models was successfully completed by implementing six supervised machine learning algorithms: Decision Tree, Random Forest, XGBoost, Logistic Regression, Support Vector Machine, and K-Nearest Neighbors.

The fifth objective of comparing model performance was achieved using Accuracy, Precision, Recall, F1-score, and Confusion Matrix as evaluation metrics.

Finally, the research identified the most influential accommodation-related factors affecting students' academic performance and wellbeing through feature importance analysis.

6.3 Research Contributions

This study contributes to educational data mining and machine learning applications in several ways.

First, it presents one of the few machine learning-based investigations examining the relationship between hall accommodation and student outcomes within the context of Noakhali Science and Technology University.

Second, unlike many previous studies focusing solely on academic performance, this research simultaneously investigated both academic achievement and student wellbeing, providing a more comprehensive understanding of accommodation-related challenges.

Third, the study demonstrates a complete end-to-end machine learning workflow beginning with primary data collection and ending with model evaluation. This workflow may serve as a useful reference for future educational data mining projects conducted in Bangladeshi universities.

Finally, the research highlights the importance of preprocessing techniques, feature engineering, and target transformation in improving machine learning performance.

7. RECOMMENDATIONS

Based on the findings and discussion presented in the previous chapters, several recommendations are proposed to improve student accommodation, academic performance, and overall wellbeing at Noakhali Science and Technology University (NSTU). These recommendations are directed toward university administrators, policymakers, researchers, and students. They are intended to address the challenges identified through the machine learning analysis and to support future improvements in residential facilities and educational planning.

7.1 Recommendations for University Administration

The findings of this study indicate that accommodation-related issues influence students' academic experiences and wellbeing. Therefore, the university administration may consider implementing the following measures:

7.1.1 Expand Hall Accommodation Capacity

One of the primary challenges identified in this research is the limited availability of university hall seats. Many students are compelled to live in rented accommodations located far from the university campus.

Increasing the number of residential halls or expanding existing facilities would enable more students to reside on campus, reducing commuting time and transportation costs while providing greater access to academic resources and campus activities.

7.1.2 Improve Transportation Facilities

Students living outside the campus often experience long commuting times and transportation-related challenges.

The university may collaborate with local transportation providers or introduce shuttle bus services during peak class hours to improve accessibility and reduce commuting difficulties.

Improved transportation may contribute to higher attendance rates and better academic performance.

7.1.3 Enhance Internet Facilities

Internet availability emerged as one of the important predictors in the machine learning models.

Providing reliable and high-speed internet access within university halls and common study areas would support online learning, research activities, assignment submission, and communication with instructors.

7.1.4 Develop Better Study Environments

Students require quiet, comfortable, and well-equipped study spaces.

The university should increase the availability of reading rooms, discussion areas, computer laboratories, and library facilities to promote effective learning among both resident and non-resident students.

7.1.5 Improve Student Safety

Safety was identified as an influential factor affecting student wellbeing.

The university should strengthen campus security by improving lighting, installing CCTV cameras, increasing security personnel, and enhancing emergency response systems within residential areas.

7.1.6 Strengthen Student Support Services

Students experiencing financial stress, academic pressure, or psychological difficulties should have access to counseling and support services.

The university may establish dedicated counseling centers, academic advising programs, and peer mentoring initiatives to promote student wellbeing.

7.2 Recommendations for Students

Students also play an important role in improving their academic performance and overall wellbeing. Based on the findings of this study, students are encouraged to:

- Maintain regular class attendance.
- Develop effective study habits.
- Manage their daily schedules efficiently.
- Ensure adequate sleep and physical well-being.
- Utilize university learning resources.
- Participate in extracurricular activities.
- Seek academic or psychological support whenever necessary.

Developing healthy academic and personal habits can positively influence both educational achievement and overall life satisfaction.

7.3 Recommendations for Policymakers

Educational policymakers may consider the findings of this research when planning future investments in higher education infrastructure. The following actions are recommended:

- Increase funding for university residential facilities.
- Support transportation improvements around university campuses.

- Invest in digital learning infrastructure.
- Promote student-centered welfare policies.
- Encourage the integration of educational data analytics into institutional decision-making.

These initiatives could contribute to improving educational quality across public universities in Bangladesh.

7.4 Recommendations for Future Researchers

Although this research successfully demonstrated the application of machine learning for predicting student outcomes, several opportunities remain for future studies. Future researchers are encouraged to:

- Collect larger datasets involving several hundred or several thousand students.
- Include students from multiple universities throughout Bangladesh.
- Incorporate additional variables such as family income, parental education, scholarship status, semester GPA, and extracurricular achievements.
- Compare deep learning models with traditional machine learning algorithms.
- Apply cross-validation and advanced hyperparameter optimization techniques.
- Investigate longitudinal datasets to examine changes in student wellbeing over time.
- Explore explainable artificial intelligence (XAI) methods such as SHAP and LIME to improve model interpretability.

Such improvements may increase predictive accuracy while providing deeper insights into student success.

7.5 Overall Recommendation

This research demonstrates that machine learning can effectively identify factors associated with students' academic performance and wellbeing. Universities should consider integrating data-driven approaches into educational planning and policy development.

Rather than relying solely on intuition or descriptive statistics, institutional decision-makers can use predictive analytics to identify at-risk students, evaluate accommodation policies, and allocate resources more effectively.

Expanding residential facilities, improving transportation, strengthening internet accessibility, enhancing student safety, and promoting supportive learning environments are likely to contribute to improved academic performance and student wellbeing.

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